PROJECT REPORT

OLYMPICS GAMES ANALYTICS

Medal Trends and Country Performance

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Tool Used: Tableau
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A data visualization project based on Olympic country medal records

Declaration

I hereby submit this project report titled “Olympics Games Analytics: Medal Trends and Country Performance”. The report describes the problem, dataset structure, preparation approach, Tableau worksheets, dashboards, observations, limitations, and possible future improvements. The analysis is based on the workbook Olympics_Games_Analytics.twbx supplied for this project.

Student Signature: _____ Date: _____

Acknowledgement

I would like to express my sincere gratitude to my university and faculty for providing the opportunity to work on this data visualization project. I also acknowledge the developers of Tableau and the providers of the Olympic medal dataset used in the workbook. This project helped me understand how structured data can be explored and communicated through charts and dashboards.

Abstract

This project uses Tableau to explore Olympic medal records and compare country-level performance. The workbook contains one data source labelled “Country Medals (Olympic_Project_Data)” and uses the fields Country, Year, Gold, Silver, Bronze, Total, and a calculated measure labelled Gold Rank Selected. Nine worksheets examine total medals, medal categories, and year-wise trends. Two dashboards combine the visualizations into consolidated views. The project demonstrates a practical workflow for turning a tabular dataset into visual summaries that are easier to compare and interpret. Exact medal totals and rankings should be read from the live workbook; this report does not invent numerical findings that are not independently visible in the workbook extract.

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1. Introduction

The Olympic Games bring together athletes representing countries from around the world. Medal records provide one way to study and compare sporting achievements. However, a table containing many countries, years, and medal categories can be difficult to interpret quickly. Data visualization helps present this information in a more accessible form.

The project, “Olympics Games Analytics: Medal Trends and Country Performance”, uses Tableau to organize country medal data into worksheets and dashboards. The analysis focuses on total medals and the three medal categories—gold, silver, and bronze—along with year-wise comparisons.

2. Problem Statement

The main problem addressed in this project is how to make Olympic medal records easier to explore and compare. A plain dataset may contain many rows and numerical fields, making it difficult to identify country-level differences or changes across years. The project addresses this problem by creating visual comparisons and consolidated dashboards in Tableau.

3. Objectives and Scope

- Compare the total medal counts recorded for different countries.
- Study gold, silver, and bronze medal measures separately.
- Examine how medal totals and medal categories vary by year.
- Present related charts together through Tableau dashboards.
- Demonstrate a repeatable workflow for data preparation and visual analysis.

Scope: The current workbook is centred on country-level medal measures and year-wise medal trends. It does not, in its present form, provide a verified analysis of individual athletes, sport-wise performance, investment, or regional efficiency.

4. Data Collection and Dataset Description

The Tableau packaged workbook uses an embedded data extract. Its data source is labelled “Country Medals (Olympic_Project_Data)”. The source file available for inspection is Olympics_Games_Analytics.twbx. The workbook's visible data model identifies the following fields:

Field	Data role / type	Purpose in analysis
Country	Text dimension	Identifies the country represented in a record.
Year	Integer dimension	Supports year-wise comparisons.
Gold	Integer measure	Number of gold medals.
Silver	Integer measure	Number of silver medals.
Bronze	Integer measure	Number of bronze medals.
Total	Integer measure	Overall medal total used for country and year comparisons.
Gold Rank Selected	Calculated integer measure	A workbook calculation/selection field associated with gold ranking.

Data provenance note: The packaged workbook identifies the data source by the caption above, but the original public download URL, publisher, collection date, and detailed inclusion criteria are not recorded in the workbook

metadata inspected for this report. These details should be added if they are available from the instructor or original dataset provider.

5. Data Cleaning and Preparation

The following preparation checks form the recommended workflow before interpreting the visualizations. The workbook metadata confirms the field names and data types, but it does not provide an audit log proving that every check below was performed on the source file. Therefore, these are documented as preparation and validation steps rather than as claims that errors were found or corrected.

1. Confirm that Country is treated as a text dimension and that Year is stored as an integer.
2. Confirm that Gold, Silver, Bronze, and Total are numeric measures so that Tableau can aggregate them.
3. Check for blank country names, missing years, and null medal values before analysis.
4. Check whether country labels are consistent (for example, spelling and naming conventions) across records.
5. Check for duplicate country-year records and determine whether they should be combined or retained according to the dataset definition.
6. Validate the Total field against Gold + Silver + Bronze where the dataset's counting rules make that comparison appropriate.
7. Review unusual values and confirm that filters or aggregations do not unintentionally exclude records.

In Tableau, dimensions such as Country and Year can be used to group records, while medal fields can be aggregated using SUM. Any data corrections should be made in the source or through documented Tableau data-preparation steps, then rechecked in the resulting charts.

6. Tools and Methodology

Primary tool: Tableau Desktop / Tableau workbook (.twbx).

Method followed:

8. Open the packaged workbook and connect to its embedded Olympic country-medal extract.
9. Review the available dimensions and measures in the data pane.
10. Create worksheets by placing Country or Year on the view and aggregating the relevant medal measure.
11. Use separate views for total medals and individual medal categories.
12. Combine related worksheets into dashboards for a consolidated presentation.
13. Review chart labels, titles, sorting, and the displayed values before drawing conclusions.

7. Worksheet and Visualization Details

The workbook contains nine worksheets. The internal worksheet tabs are named Sheet 1 through Sheet 9. The following descriptions are based on the fields and row/column configurations stored in the workbook.

Worksheet	Fields / view configuration	Purpose
Sheet 1	Country vs SUM(Total)	Compares total Olympic medals by country.
Sheet 2	Country vs SUM(Bronze) + SUM(Gold)	Compares a combined gold-and-bronze measure by country.
Sheet 3	Country and Measure Names / Multiple Values	Displays multiple medal measures together for country comparison; the visible dependencies include Bronze and Silver.
Sheet 4	Country vs SUM(Bronze)	Compares bronze medal counts by country.
Sheet 5	Year vs SUM(Total)	Shows total medal counts across years.

Sheet 6	Year vs SUM(Gold)	Shows gold medal counts across years.
Sheet 7	Year vs SUM(Silver)	Shows silver medal counts across years.
Sheet 8	Year vs SUM(Bronze)	Shows bronze medal counts across years.
Sheet 9	Country vs Multiple Values	Combines multiple measures for country-level comparison; includes Gold, Silver, Bronze, Total and a gold-rank calculation.

Note: Sheet 2 is specifically configured with Bronze plus Gold; it should not be described as a silver-medal sheet. Sheet 3 and Sheet 9 use multiple-measure configurations. The exact chart appearance and any interactive filter behaviour should be confirmed by opening the workbook in Tableau.

8. Dashboard Development

Two dashboards are saved in the workbook: “Dashboard 2” and “Olyampic Performance Dashboard” (spelling as stored in the workbook). Dashboards bring multiple worksheet views into one screen so that users can compare related information without navigating between individual sheets.

The first dashboard can be used as a consolidated view of the medal analysis, while the second dashboard provides an additional presentation layout. During the final review, dashboard titles, legends, labels, layout alignment, and any filters should be checked for clarity and consistency.

The workbook metadata inspected for this report does not contain a separate Tableau Story object. If a Story is required by the project template, it can be created in Tableau by adding story points that guide the viewer from total medals, to medal categories, to year-wise trends, and finally to the dashboards.

9. Results and Observations

The workbook provides a visual structure for examining Olympic medal performance from several perspectives. The following are supported analytical observations about the design of the analysis; numerical rankings and specific country/year winners should be read directly from the live charts before submission.

- Country-level comparison: Sheet 1 makes it possible to compare countries using the Total measure.
- Medal composition: Separate and combined medal views allow the viewer to distinguish overall performance from individual medal categories.
- Time-based comparison: Sheets 5–8 organize medal measures by Year, allowing changes between recorded years to be examined.
- Multi-measure comparison: Sheets 3 and 9 provide a way to view more than one medal indicator in the same country-level analysis.
- Dashboard usability: The two dashboards consolidate worksheet views and make the analysis easier to present.

Before final submission, record at least two or three concrete findings from the workbook itself—for example, the country with the highest displayed total, the year with the highest displayed total, and a comparison between the medal categories. Include the exact values and the relevant sheet/dashboard name.

10. Limitations

- The original dataset source URL and collection methodology are not available in the workbook metadata inspected.

- The report does not claim specific country rankings or medal totals because those values have not been independently extracted and verified here.
- The current workbook focuses on country medal totals and year-wise medal measures; it does not establish causes behind medal performance.
- Differences in Olympic events, participation, historical country naming, and dataset counting rules may affect comparisons.
- A separate Story object was not found in the saved workbook structure.

11. Conclusion

The Olympics Games Analytics project demonstrates how Tableau can be used to organize and communicate country-level Olympic medal data. Through nine worksheets and two dashboards, the workbook supports comparisons of total medals, individual medal categories, and year-wise medal measures. The project shows how visual analysis can make tabular data easier to explore and present. It also highlights the importance of checking data types, consistency, totals, and source information before drawing conclusions.

12. Future Scope

- Add verified athlete-level and sport-wise records to study performance beyond country medal totals.
- Add filters for year, country, medal category, and Olympic edition where the source data supports them.
- Create a Tableau Story with sequential story points explaining the analysis.
- Add a data dictionary and document the source URL, download date, and data-cleaning decisions.
- Include validated key performance indicators and clearly labelled top-country or top-year summaries.
- Compare Summer and Winter Games separately if the dataset contains reliable edition information.

13. References

1. Tableau packaged workbook supplied for the project: Olympics_Games_Analytics.twbx.

2. Tableau Help documentation: https://help.tableau.com/ (for worksheet, dashboard, and data visualization concepts).

3. Original Olympic dataset provider: To be completed using the source link supplied with the dataset. The workbook metadata inspected for this report does not identify a public source URL.

Appendix A — Final Submission Checklist

- ☐ Insert the exact dataset source and access date.
- ☐ Confirm the exact medal values and top country/year observations from Tableau.
- ☐ Add screenshots of the nine worksheets and both dashboards if required by the template.
- ☐ Verify the dashboard titles and correct any spelling or formatting issues.
- ☐ If a Story is required, create it in Tableau and add its screenshots to this report.
- ☐ Replace any blank institution, faculty, course, or submission-date fields required by the official format.